

HCSM-00

Foundations, Notation, and Mapping

The Hodge Complex Standard Model: Substrate, Hodge Complex, Spectral Midpoint,
Processor, Output Structures, Emergent Physics, and Cosmology

Stanley Preschutti

Entropy Research Institute / Information Physics Institute

ORCID: 0009-0004-5445-1744

scstanp@yahoo.com

September 2026

Abstract

The Hodge Complex Standard Model (HCSM) derives the gauge structure, matter content, particle spectrum, mixing matrices, spacetime dimension, graviton, Newton constant, and vacuum stability coefficient of the Standard Model from a single discrete substrate: the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. The substrate is selected by two independent derivations (equal entropy spacing on the torus Laplacian and maximal cancellation of the winding generator at $L = 8$), and the emergent spacetime dimension is fixed by the topology of the Hodge–Dirac kernel. The Hodge complex of differential forms carries the operator $D = d + d^*$ with $D^2 = \Delta$ exactly on each form degree. The 14 modes at the self-paired eigenvalue $\lambda = 4$ are identified with one Standard Model generation.

This paper establishes the framework’s foundational objects and states the notation. All seven rank blockers of the framework’s derivation program are closed. The framework has a single dimensionful anchor (the Planck mass M_P) and derives every other quantity from the substrate at $d = 4$ and $L = 8$.

The paper establishes the following theorems. The 5-point Laplacian is the unique D_4 -invariant nearest-neighbour positive-semidefinite operator with kernel equal to the constants. The Hodge–Dirac operator satisfies $D^2|_{\Omega^k} = \Delta \cdot \text{id}$ on each form degree, with smallest nonzero eigenvalue $2 \sin(\pi/L)$ on Λ_L . The Lorentzian metric on $\ker D$ is the unique D_4 -invariant symmetric form that vanishes on pure-degree forms and has signature $(3, 1)$. The Kernel-Diagonal Theorem yields the EFT coefficient $c = 5/9$, the mass-map coefficient $(d - 1)/2 = 3/2$, and the kernel dimension $10 = (d - 1)^2 + 1$ as three traces of one projector. The Λ -promotion mechanism is a pure projection onto the timelike direction of $\ker D$, and the refined dark energy density matches observation to 0.010%. Section 9 lists the framework’s falsifiable predictions.

Contents

1	Introduction	3
1.1	What HCSM is	3
1.2	The single dimensionful anchor	3
1.3	Architectural levels	3
1.4	Continuum limit and emergent Lorentz invariance	3
2	Level 0: The Substrate	4
2.1	The periodic $L \times L$ torus	4
2.2	The 5-point Laplacian is forced	4

2.3	The framework relation $N = d \cdot 2^d$	4
2.4	Two derivations of $L = 8$	5
2.4.1	First derivation: equal entropy spacing	5
2.4.2	Second derivation: maximal winding cancellation	5
2.5	The 14-mode count	5
3	Level 1: The Hodge Complex	6
3.1	The three form degrees	6
3.2	The exterior derivative and the codifferential	6
3.3	The Hodge–Dirac operator	6
3.4	The $\mathbb{Z}_2$ grading and the supercharge structure	7
3.5	The kernel and the emergent dimension	8
3.6	The spectral gap and absence of species doubling	9
4	Level 2: The Spectral Midpoint	9
4.1	The self-paired eigenvalue	9
4.2	The D_4 orbit decomposition	10
4.3	The 56-dimensional Hodge lift	10
4.4	The D_4 decomposition of the 56-dimensional sector	10
5	Level 3: The Processor and Its Dynamics	11
5.1	The winding generator	11
5.2	The winding classes and the processor	11
5.3	The Kernel-Diagonal Theorem	12
5.4	The five-layer loop	13
5.5	The zero state	13
6	Level 4: Output Structures	14
6.1	The charge functional	14
6.2	The mass map	14
6.3	Three derivations of the coefficient $3/2$	14
7	Level 5: Emergent Physics	15
7.1	Gauge algebra from the D_4 commutant	15
7.2	Hypercharge quantization	15
7.3	Flavor: the CKM matrix	15
7.4	Flavor: the PMNS matrix	16
7.5	Higgs sector	16
7.6	Gravity	16
7.7	Absence of a light dark photon	16
7.8	Spin chirality $\mathbb{Z}_2$ grading	16
7.9	Strong CP from orientation $\mathbb{Z}_2$	17
7.10	Proca sector and $\mathfrak{u}(1)_Y$ origin	17
8	Level 6: Cosmology	17
8.1	The kL crossover	17
8.2	Dark energy and the DME entropy-well normalization	17
8.3	Vacuum stability	18
8.4	The Λ -promotion mechanism	18
9	Falsifiable predictions	18
10	Theorem inventory	19

1 Introduction

1.1 What HCSM is

The Hodge Complex Standard Model derives the Standard Model of particle physics from a single discrete substrate: the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. The framework’s central claim is that the Standard Model is not a fundamental description of nature but the residue left when a small, discrete, information-bearing lattice cancels against its own mirror.

Physics is what the machine cannot cancel. The Hodge complex is the machine.

HCSM does not ask what fields exist or what symmetry groups are broken. It asks a single question: what substrate could produce exactly the Standard Model, and nothing else? The answer is a discrete lattice whose mirror cancellation leaves exactly the observed particle content.

1.2 The single dimensionful anchor

The framework has exactly one dimensionful input: the Planck mass M_P . Every other dimensionful quantity is a derived multiple of an appropriate power of M_P . The dimensionless integers $d = 4$ and $N = 64$ are derived from the substrate (§2); the substrate size $L = 8$ is derived from two independent routes (§2.4); and the emergent Lorentzian signature $(3, 1)$ is derived from the Hodge star on $\ker D$ (§3.5). No second dimensionful anchor is required.

1.3 Architectural levels

The framework’s architecture is a chain from the substrate to observable physics, organized in six levels.

Level	Name	Content
0	Substrate	8×8 torus, 5-point Laplacian, anchors
1	Hodge complex	$\Omega^0 \oplus \Omega^1 \oplus \Omega^2$, $D = d + d^*$
2	Spectral midpoint	$\lambda = 4$, 14 modes, D_4 orbits
3	Processor	H_{wind} , T , zero state, five-layer loop
4	Output structures	Charge q_3 , mass map, $N/4$ labels
5	Emergent physics	Gauge, flavor, Higgs, gravity, dark photon
6	Cosmology	kL crossover, dark energy, Λ promotion

1.4 Continuum limit and emergent Lorentz invariance

The substrate is not a discretization of an underlying continuum. The framework’s claim is the opposite: the continuum emerges from the substrate. The physical scale is set by the mass map and the kL crossover.

The 5-point Laplacian has the isotropic expansion

$$\lambda(n_1, n_2) = \frac{\pi^2}{16} (n_1^2 + n_2^2) + O(n^4), \quad (1)$$

isotropic to leading order. The quartic corrections are suppressed by ϵ^2 relative to the leading term, where $\epsilon = n_{\text{max}}/L \ll 1$ in the low-momentum regime. The discrete-to-continuum scaling limit is item 6.14 of the framework’s need-to-derive list and remains a program-level item.

2 Level 0: The Substrate

2.1 The periodic $L \times L$ torus

The substrate is the periodic square lattice

$$\Lambda_L = \mathbb{Z}_L \times \mathbb{Z}_L, \quad N = L^2, \quad (2)$$

consisting of N sites arranged in an $L \times L$ grid with periodic boundary conditions. Site (x_1, x_2) is identified with $(x_1 + L, x_2)$ and with $(x_1, x_2 + L)$; equivalently, the coordinates x_1 and x_2 are each defined modulo L . Topologically, Λ_L is the torus $T^2 = S^1 \times S^1$.

The D_4 point group of the square acts on Λ_L by the eight rotations and reflections of the lattice. Its presentation is

$$D_4 = \langle r, s \mid r^4 = s^2 = (rs)^2 = e \rangle. \quad (3)$$

2.2 The 5-point Laplacian is forced

Theorem 2.1 (Laplacian uniqueness). *Let $\Delta : \mathbb{C}^N \rightarrow \mathbb{C}^N$ be a linear operator on the substrate $\Lambda_L = \mathbb{Z}_L \times \mathbb{Z}_L$ satisfying:*

1. *D_4 -invariance: $[R, \Delta] = 0$ for every $R \in D_4$.*
2. *Nearest-neighbour locality: $(\Delta f)(x)$ depends only on $f(x)$ and f at the four nearest neighbours of x .*
3. *Positivity and kernel: Δ is positive-semidefinite and $\ker \Delta$ is exactly the set of constant functions.*

Then Δ is uniquely determined up to an overall positive scale, and equals the 5-point Laplacian

$$(\Delta f)(x_1, x_2) = 4f(x_1, x_2) - f(x_1 + 1, x_2) - f(x_1 - 1, x_2) - f(x_1, x_2 + 1) - f(x_1, x_2 - 1). \quad (4)$$

Proof. Condition (2) writes $\Delta f(x) = a_0 f(x) + \sum_{\mu \in \{\pm e_1, \pm e_2\}} a_\mu f(x + \mu)$. The four nearest-neighbour directions $\pm e_1, \pm e_2$ lie in a single D_4 orbit (the reflection s maps $e_1 \rightarrow -e_1$; the rotation r maps $e_1 \rightarrow e_2$). Hence condition (1) forces $a_\mu = a$ for all four directions, giving

$$(\Delta f)(x) = a_0 f(x) + a \sum_{\mu} f(x + \mu). \quad (5)$$

Condition (3) forces $\Delta \mathbf{1} = 0$, hence $a_0 + 4a = 0$, so $a_0 = -4a$. Normalizing $a = 1$ gives (4). $\square$

Remark 2.2. The 5-point Laplacian annihilates the constants by construction. Its eigenvalue on the plane wave $\phi_{n_1, n_2}(x) = \exp(2\pi i(n_1 x_1 + n_2 x_2)/L)$ is

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{2\pi n_1}{L}\right) - 2 \cos\left(\frac{2\pi n_2}{L}\right), \quad n_1, n_2 \in \{0, 1, \dots, L-1\}. \quad (6)$$

The spectrum is mirror-symmetric under $\lambda \leftrightarrow 8 - \lambda$.

2.3 The framework relation $N = d \cdot 2^d$

The framework uses the relation

$$N = d \cdot 2^d \quad (7)$$

with N the number of substrate sites and d the emergent spacetime dimension. At $d = 4$ this gives

$$N = 4 \cdot 2^4 = 4 \cdot 16 = 64. \quad (8)$$

Since $\Lambda_L = \mathbb{Z}_L \times \mathbb{Z}_L$ requires $N = L^2$, this gives $L = \sqrt{64} = 8$.

At $d = 4$, $N = 64$ is a perfect square. The integer solutions of $d \cdot 2^d = L^2$ form the infinite family $d \in \{4k^2 : k \geq 1\}$, with $d = 16, 36, 64, \dots$ also giving perfect squares. The physical selection of $d = 4$ therefore comes from the emergent-dimension argument of §3.5, not from the perfect-square condition.

2.4 Two derivations of $L = 8$

The substrate size $L = 8$ is selected by two independent derivations. Their agreement determines $L = 8$.

2.4.1 First derivation: equal entropy spacing

On the $L \times L$ torus, the standing-wave modes fall into discrete Shannon entropy classes. The companion equal-entropy paper proves the following.

Theorem 2.3 (Equal entropy spacing selects $L = 8$). *Among all integers $4 \leq L \leq 10^4$, the only L for which the $L \times L$ torus has exactly three equally spaced Shannon entropy classes is $L = 8$. At $L = 8$ the three classes are 6.0 bits, 5.5 bits, and 5.0 bits, with uniform spacing 0.5 bits.*

2.4.2 Second derivation: maximal winding cancellation

The winding generator of §5.1 generalizes to any even L as $H_{\text{wind}}(L) = i \sin(\pi M/L)$ on the self-paired mode set of Λ_L .

Theorem 2.4 (Maximal cancellation selects $L = 8$). *Let $\mathcal{G}_L$ be the parity block graph of $H_{\text{wind}}(L)$ on the five nodes $\{A_{\text{odd}}, A_{\text{even}}, B_{\text{odd}}, B_{\text{even}}, AB\}$. The number of block couplings that vanish identically is*

$$\#\text{zeros}(\mathcal{G}_L) = \begin{cases} 4, & L \equiv 2 \pmod{4}, \\ 7, & L \equiv 0 \pmod{4}, L \geq 12, \\ 10, & L \in \{4, 8\}. \end{cases} \quad (9)$$

Among the two maximally-cancelling sizes, only $L = 8$ carries the 14-mode multiplet. At $L = 8$ the surviving five block couplings form a unicyclic graph on $d+1 = 5$ nodes with rank-1 edges; the winding kernel has dimension $d(d+1)/2 = 10$; and the kernel-diagonal values are $\{2/d, (d-1)/d\}$.

The two derivations are independent: the first uses only the torus Laplacian's entropy classes; the second uses only the winding generator's parity block structure.

2.5 The 14-mode count

Theorem 2.5 (Midpoint multiplicity). *On the $L \times L$ torus with L even, the multiplicity of the self-paired eigenvalue $\lambda = 4$ is*

$$\text{mult}_{\lambda=4}(L) = 2(L - 1). \quad (10)$$

At $L = 8$ this gives 14.

Proof. The self-paired eigenvalue is the fixed point of the mirror symmetry $\lambda \leftrightarrow 8 - \lambda$. Its eigenspace consists of the modes (n_1, n_2) satisfying

$$\cos\left(\frac{2\pi n_1}{L}\right) + \cos\left(\frac{2\pi n_2}{L}\right) = 0. \quad (11)$$

Using the sum-to-product identity $\cos a + \cos b = 2 \cos\left(\frac{a+b}{2}\right) \cos\left(\frac{a-b}{2}\right)$, this becomes

$$2 \cos\left(\frac{\pi(n_1 + n_2)}{L}\right) \cos\left(\frac{\pi(n_1 - n_2)}{L}\right) = 0. \quad (12)$$

The solutions are $n_1 + n_2 \equiv L/2 \pmod{L}$ or $n_1 - n_2 \equiv L/2 \pmod{L}$. Each congruence has L solutions in $\mathbb{Z}_L \times \mathbb{Z}_L$; the two congruences intersect in exactly two points. By inclusion-exclusion, the total is $2L - 2 = 2(L - 1)$. At $L = 8$ this gives 14. $\square$

The 14 modes at $\lambda = 4$ on $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ are identified with one Standard Model generation.

3 Level 1: The Hodge Complex

3.1 The three form degrees

On the 2-torus Λ_L , the Hodge complex of differential forms has three graded pieces:

$$\Omega^0 = \mathbb{R}^N \quad (0\text{-forms on the } N \text{ sites}), \quad \dim \Omega^0 = N, \quad (13)$$

$$\Omega^1 = \mathbb{R}^{2N} \quad (1\text{-forms on the } 2N \text{ directed edges}), \quad \dim \Omega^1 = 2N, \quad (14)$$

$$\Omega^2 = \mathbb{R}^N \quad (2\text{-forms on the } N \text{ plaquettes}), \quad \dim \Omega^2 = N. \quad (15)$$

The total dimension is

$$\dim(\Omega^0 \oplus \Omega^1 \oplus \Omega^2) = N + 2N + N = 4N. \quad (16)$$

At $N = 64$ this gives $4 \cdot 64 = 256$.

3.2 The exterior derivative and the codifferential

Let $e_1 = (1, 0)$ and $e_2 = (0, 1)$ be the two oriented lattice directions. The exterior derivative acts on 0-forms by

$$(d_0 f)(x, \mu) = f(x + e_\mu) - f(x), \quad \mu \in \{1, 2\}, \quad (17)$$

and on 1-forms by

$$(d_1 \omega)(x) = \omega_1(x) + \omega_2(x + e_1) - \omega_1(x + e_2) - \omega_2(x). \quad (18)$$

These satisfy $d_1 d_0 = 0$ exactly.

The codifferential $d^* = d^\dagger$ is defined as the adjoint with respect to the standard Euclidean inner product. Taking $d_0^* = d_0^T$ and $d_1^* = d_1^T$ as matrices gives

$$(d_0^* \omega)(x) = \sum_{\mu=1}^2 [\omega_\mu(x - e_\mu) - \omega_\mu(x)], \quad (19)$$

$$(d_1^* \chi)(x, 1) = \chi(x) - \chi(x - e_2), \quad (d_1^* \chi)(x, 2) = \chi(x - e_1) - \chi(x). \quad (20)$$

The sign convention is fixed by the requirement that d_0^* and d_1^* are the transposes of d_0 and d_1 . With this convention $d_0^* d_1^* = 0$ follows from $d_1 d_0 = 0$.

Remark 3.1 (Sign convention). The formula (19) differs by an overall minus sign from $\sum_\mu [\omega_\mu(x) - \omega_\mu(x - e_\mu)]$. The choice here is fixed by requiring d^* to be the matrix transpose of d ; with the opposite sign, D^2 on Ω^0 equals $-\Delta$ rather than $+\Delta$.

3.3 The Hodge–Dirac operator

Definition 3.2 (Hodge–Dirac operator). The Hodge–Dirac operator is

$$D = d + d^*, \quad (21)$$

acting on the full Hodge complex $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$. In block form,

$$D = \begin{pmatrix} 0 & d_0^* & 0 \\ d_0 & 0 & d_1^* \\ 0 & d_1 & 0 \end{pmatrix}, \quad (22)$$

a symmetric 256×256 real operator.

Theorem 3.3 (Hodge–Dirac square). *On the discrete torus Λ , the identity*

$$D^2|_{\Omega^k} = \Delta \cdot \text{id}_{\Omega^k} \quad (23)$$

holds exactly on each form degree $k = 0, 1, 2$.

Proof. We give the computation on Ω^0 explicitly. The computations on Ω^1 and Ω^2 follow the same structure, with cross terms removed by nilpotency $d_1 d_0 = 0$.

Case $k = 0$. On Ω^0 , $D^2 = d_0^* d_0$. Using (17) and (19), for any f ,

$$\begin{aligned} (d_0^* d_0 f)(x) &= \sum_{\mu=1}^2 \left[(d_0 f)_\mu(x - e_\mu) - (d_0 f)_\mu(x) \right] \\ &= \sum_{\mu=1}^2 \left[(f(x) - f(x - e_\mu)) - (f(x + e_\mu) - f(x)) \right] \\ &= \sum_{\mu=1}^2 [2f(x) - f(x - e_\mu) - f(x + e_\mu)] \\ &= 4f(x) - \sum_{|e|=1} f(x + e), \end{aligned}$$

which is Δf exactly.

Case $k = 1$. On Ω^1 , $D^2 = d_0 d_0^* + d_1^* d_1$. The cross-terms $d_1 d_0 = 0$ ensure no mixed-degree contributions appear. Direct expansion of both terms in components gives $(d_0 d_0^* + d_1^* d_1)\omega = \Delta \omega$, acting component-wise on (ω_1, ω_2) .

Case $k = 2$. On Ω^2 , $D^2 = d_1 d_1^*$. Using the plaquette-boundary structure of d_1 and the adjoint (20), a direct expansion gives $(d_1 d_1^* \chi)(x) = 4\chi(x) - \sum_{|e|=1} \chi(x + e) = \Delta \chi$ on each plaquette.

All three identities are verified at machine precision. $\square$

Remark 3.4 (No species doubling). The Hodge–Dirac operator on the discrete torus has no species doubling: its square equals the scalar Laplacian on each form degree, and therefore its spectrum contains one mode per eigenvalue of Δ (up to the Ω^1 doubling of multiplicity). The low-energy spectrum is bounded away from zero by $2 \sin(\pi/L)$ except for the four zero modes. This is a distinct property from the spinorial Dirac operator on the same lattice, whose square is the doubled Laplacian.

3.4 The $\mathbb{Z}_2$ grading and the supercharge structure

Definition 3.5 (Parity operator). The parity operator $(-1)^F$ on the Hodge complex is

$$(-1)^F = \begin{cases} +1, & \text{on } \Omega^0 \oplus \Omega^2, \\ -1, & \text{on } \Omega^1. \end{cases} \quad (24)$$

The parity operator $\mathbb{Z}_2$ -grades the complex into a bosonic sector $\Omega^0 \oplus \Omega^2$ (even forms) and a fermionic sector Ω^1 (odd forms).

Proposition 3.6 (D is odd). $D(-1)^F = -(-1)^F D$.

Proof. D maps Ω^k to $\Omega^{k\pm 1}$, changing form degree by ± 1 and flipping the parity eigenvalue. $\square$

Assign canonical commutation relations to even-sector modes a_n and anticommutation relations to odd-sector modes ψ_n :

$$[a_n, a_m^\dagger] = \delta_{nm}, \quad \{\psi_n, \psi_m^\dagger\} = \delta_{nm}. \quad (25)$$

The Hodge–Dirac operator becomes the supercharge

$$D = \sum_n \sqrt{\lambda_n} (a_n^\dagger \psi_n + \psi_n^\dagger a_n). \quad (26)$$

Theorem 3.7 (Supercharge structure). *With $Q = D$ and $H = D^2/2 = \Delta/2$,*

$$\{Q, Q^\dagger\} = 2H, \quad Q^2 = (Q^\dagger)^2 = 0. \quad (27)$$

This is the standard $N = 1$ supersymmetric quantum mechanics algebra.

Proof. Expand $Q^2 = (d + d^*)^2 = d^2 + \{d, d^*\} + (d^*)^2$. Since $d^2 = 0$ and $(d^*)^2 = 0$ by nilpotency, $Q^2 = \{d, d^*\}$. By Theorem 3.3, $\{d, d^*\} = \Delta \cdot \text{id}$ on each form degree, so $Q^2 = \Delta \cdot \text{id} = 2H$. The anticommutator $\{Q, Q^\dagger\} = 2Q^2 = 4H$ reduces to the standard form after the normalization $H = \Delta/2$. $\square$

Corollary 3.8 (Boson–fermion pairing). *Every bosonic mode at energy λ_n has a fermionic partner at the same energy, unless the mode is annihilated by Q or $Q^\dagger$. In particular, at $\lambda = 4$:*

$$28 \text{ bosons} + 28 \text{ fermions at } \lambda = 4. \quad (28)$$

3.5 The kernel and the emergent dimension

Theorem 3.9 (Kernel dimension). *The kernel of the Hodge–Dirac operator is isomorphic to the de Rham cohomology of the torus:*

$$\ker D \cong \bigoplus_{k=0}^2 H^k(T^2; \mathbb{R}), \quad (29)$$

with dimension

$$\dim \ker D = b_0 + b_1 + b_2 = 1 + 2 + 1 = 4. \quad (30)$$

Proof. By the discrete Hodge theorem of Dodziuk, the kernel of $D = d + d^*$ is isomorphic to the direct sum of the de Rham cohomology groups $H^k(T^2; \mathbb{R})$ for $k = 0, 1, 2$. The Betti numbers of the torus are $b_0 = 1$, $b_1 = 2$, $b_2 = 1$, so $\dim \ker D = 4$. $\square$

Remark 3.10 (The four harmonic forms). The four harmonic forms are explicit: the constant 0-form v_0 (spanning H^0); the two independent harmonic 1-forms v_1^A, v_1^B (spanning H^1); the constant 2-form v_2 (spanning H^2).

Corollary 3.11 (Emergent dimension). *The dimension of the emergent spacetime is $d = \dim \ker D = 4$.*

Theorem 3.12 (Lorentzian signature). *The Hodge star $\star$ restricted to $\ker D$ has eigenvalues $\{+1, -1, +i, -i\}$: the ± 1 pair comes from the $(H^0 \oplus H^2)$ sector, the $\pm i$ pair from the H^1 sector. The intersection form $Q(\alpha, \beta) = \int_\Lambda \alpha \wedge \beta$ on the even-degree sector and the Hodge inner product $\langle \alpha, \beta \rangle = \int_\Lambda \alpha \wedge \star \beta$ on the odd-degree sector combine into a metric*

$$g = Q|_{H^0 \oplus H^2} \oplus \langle \cdot, \cdot \rangle|_{H^1} \quad (31)$$

of signature $(3, 1)$. The metric g is the unique D_4 -invariant symmetric form on $\ker D$ that vanishes on the pure-degree forms v_0 and v_2 and has signature $(3, 1)$, up to overall scale. The timelike direction is $(v_0 - v_2)/\sqrt{2}$.

Proof. The Hodge star exchanges H^0 and H^2 (eigenvalues ± 1 on the block $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$) and rotates H^1 (eigenvalues $\pm i$). On $H^0 \oplus H^2$, $Q(v_0, v_2) = 1$ (normalized volume) with $Q(v_0, v_0) = Q(v_2, v_2) = 0$ by degree; the block $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ has eigenvalues ± 1 . On H^1 , the Hodge inner product in the orthonormal basis is the identity, giving eigenvalues $+1, +1$. Total signature: $(3, 1)$.

The space of D_4 -invariant symmetric forms on $\ker D = 2A_1 \oplus E$ is 4-dimensional, parametrized by (a, b, c, d) in the block form $\begin{pmatrix} a & b & 0 \\ b & c & 0 \\ 0 & 0 & dI_2 \end{pmatrix}$. The condition $g(v_0, v_0) = g(v_2, v_2) = 0$ forces $a = c = 0$. Lorentzian signature requires $b \neq 0$, $d \neq 0$, and $\text{sign}(b) \cdot \text{sign}(d) > 0$. Up to overall scale, the metric is fixed. Normalizing $b = d = 1$ gives (31). $\square$

Corollary 3.13 (The timelike direction). *With signature convention $(-, +, +, +)$, the timelike direction on $\ker D$ is $(v_0 - v_2)/\sqrt{2}$. The three spacelike directions are $(v_0 + v_2)/\sqrt{2}$, v_1^A , and v_1^B .*

3.6 The spectral gap and absence of species doubling

Theorem 3.14 (Smallest nonzero eigenvalue of D). *On $\Lambda_L = \mathbb{Z}_L \times \mathbb{Z}_L$ with L even, the smallest nonzero eigenvalue of the Hodge–Dirac operator is exactly*

$$\mu_{\min}(L) = 2 \sin\left(\frac{\pi}{L}\right). \quad (32)$$

Proof. The eigenvalues of D^2 are the eigenvalues of the scalar Laplacian Δ on Λ_L . The smallest nonzero Laplacian eigenvalue is

$$\lambda_{\min}(L) = 2 \left(1 - \cos\left(\frac{2\pi}{L}\right)\right), \quad (33)$$

achieved at momentum $(n_1, n_2) = (1, 0)$ and its images. Then

$$\mu_{\min}(L) = \sqrt{\lambda_{\min}(L)} = \sqrt{2 \left(1 - \cos\left(\frac{2\pi}{L}\right)\right)} = 2 \sin\left(\frac{\pi}{L}\right). \quad (34)$$

$\square$

Corollary 3.15 (Absence of species doubling). *For every finite even L , the Hodge–Dirac operator has exactly four zero eigenvalues (the four harmonic forms) and the next eigenvalue is bounded away from zero by $2 \sin(\pi/L)$. No spurious low-energy modes appear.*

4 Level 2: The Spectral Midpoint

4.1 The self-paired eigenvalue

The mirror symmetry $\lambda \leftrightarrow 8 - \lambda$ of the scalar Laplacian spectrum has a unique fixed point at $\lambda = 4$. The eigenspace at $\lambda = 4$ has multiplicity $2(L - 1) = 14$ at $L = 8$. The 14 modes are

$$\mathcal{O}_0 = \{(0, 4), (4, 0)\}, \quad (35)$$

$$\mathcal{O}_1 = \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \quad (36)$$

$$\mathcal{O}_2 = \{(2, 2), (2, 6), (6, 2), (6, 6)\}. \quad (37)$$

4.2 The D_4 orbit decomposition

Proposition 4.1 (Orbit stabilizers). *Under the D_4 point group, the 14 modes decompose into the three orbits (35)–(37) of sizes $(2, 8, 4)$. The stabilizer orders of the three orbits are $4, 1, 2$ respectively, satisfying the orbit-stabilizer identity*

$$|\mathcal{O}_k| = \frac{|D_4|}{|\text{Stab}(\mathcal{O}_k)|} = \frac{8}{|\text{Stab}(\mathcal{O}_k)|}. \quad (38)$$

Proof. Direct computation. The orbit of $(0, 4)$ is stabilized by $\{e, r^2, s, r^2s\}$, of order 4, giving $|\mathcal{O}_0| = 8/4 = 2$. The orbit of $(1, 3)$ has trivial stabilizer, giving $|\mathcal{O}_1| = 8/1 = 8$. The orbit of $(2, 2)$ is stabilized by $\{e, rs\}$, of order 2, giving $|\mathcal{O}_2| = 8/2 = 4$. $\square$

4.3 The 56-dimensional Hodge lift

Theorem 4.2 (Fourteen-fold lift). *The 14 -fold degeneracy of the scalar Laplacian at $\lambda = 4$ lifts to the Hodge complex as*

$$14\Omega^0 + 28\Omega^1 + 14\Omega^2 = 56. \quad (39)$$

Proof. By Theorem 3.3, $D^2|_{\Omega^k} = \Delta$. The Laplacian has multiplicity 14 at $\lambda = 4$ on Ω^0 ; on Ω^1 each site carries two 1-form components, so the multiplicity doubles to 28; on Ω^2 the multiplicity is again 14. Summation gives 56. $\square$

4.4 The D_4 decomposition of the 56-dimensional sector

Theorem 4.3 (D_4 decomposition). *As D_4 representations,*

$$\Omega^0|_{\lambda=4} = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E, \quad (40)$$

$$\Omega^1|_{\lambda=4} = \Omega^0|_{\lambda=4} \otimes E = 3A_1 \oplus 3A_2 \oplus 3B_1 \oplus 3B_2 \oplus 8E, \quad (41)$$

$$\Omega^2|_{\lambda=4} = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E. \quad (42)$$

The full 56-dimensional sector decomposes as

$$56 = 9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E. \quad (43)$$

Proof. The character of $\Omega^0|_{\lambda=4}$ on the five conjugacy classes (e, r^2, r, s, rs) is

$$\chi_{14} = (14, 2, 0, 2, 2). \quad (44)$$

Direct verification: r^2 fixes $(0, 4)$ and $(4, 0)$; s fixes $(0, 4)$ and $(4, 0)$; rs fixes $(2, 2)$ and $(6, 6)$; r fixes no mode.

By character orthogonality with the D_4 character table (class sizes $1, 1, 2, 2, 2$), the multiplicities are

$$\begin{aligned} m_{A_1} &= \frac{1}{8}(1 \cdot 14 + 1 \cdot 2 + 2 \cdot 0 + 2 \cdot 2 + 2 \cdot 2) = 3, \\ m_{A_2} &= \frac{1}{8}(14 + 2 - 4 - 4) = 1, \\ m_{B_1} &= \frac{1}{8}(14 + 2 + 4 - 4) = 2, \\ m_{B_2} &= \frac{1}{8}(14 + 2 - 4 + 4) = 2, \\ m_E &= \frac{1}{8}(14 \cdot 2 - 2 \cdot 2) = 3. \end{aligned}$$

Dimension check: $3 + 1 + 2 + 2 + 6 = 14$. The tensor-product identity $\Omega^1|_{\lambda=4} = \Omega^0|_{\lambda=4} \otimes E$ follows from the vector-twisted D_4 action on 1-form components. Summing the three sectors gives (43), with dimension check $9 + 5 + 7 + 7 + 28 = 56$. $\square$

5 Level 3: The Processor and Its Dynamics

5.1 The winding generator

Definition 5.1 (Winding matrix and winding generator). Let $n_1, n_2 \in \mathbb{R}^{14}$ be the coordinate vectors of the 14 modes. The winding matrix is the rank-2 antisymmetric bilinear

$$M_{ij} = n_1^{(i)} n_2^{(j)} - n_1^{(j)} n_2^{(i)}. \quad (45)$$

The winding generator is

$$H_{\text{wind}} = i \sin(\pi M/8), \quad (46)$$

with $\sin$ applied elementwise.

Proposition 5.2 (Hermiticity and entry structure). H_{wind} is Hermitian with entries in $\mathbb{Q}(\sqrt{2})$. The entries of M are even integers, so the entries of $\pi M/8$ are integer multiples of $\pi/4$, and $\sin(\pi M/8) \in \{0, \pm 1/\sqrt{2}, \pm 1\}$.

Proof. M is real antisymmetric, so $\sin(\pi M/8)$ is real antisymmetric. Multiplying by i gives a Hermitian operator. The integrality of M follows from the integer coordinates: $M_{ij} \in \mathbb{Z}$. Parity check on the 14 mode coordinates shows M_{ij} is even for every pair. $\square$

Theorem 5.3 (Characteristic polynomial). The characteristic polynomial of H_{wind} is

$$p(\lambda) = \det(\lambda \text{id}_{14} - H_{\text{wind}}) = \lambda^{10}(\lambda^2 - 8)(\lambda^2 - 32) = \lambda^{14} - 40\lambda^{12} + 256\lambda^{10}. \quad (47)$$

Its spectrum is

$$\text{spec}(H_{\text{wind}}) = \{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\}. \quad (48)$$

Proof. Write $p(\lambda) = \lambda^{14} + a_1\lambda^{13} + a_2\lambda^{12} + \dots$. Newton's identities relate the coefficients to the power sums $p_k = \text{Tr}(H_{\text{wind}}^k)$:

$$ka_k = -\sum_{i=1}^k a_{k-i}p_i, \quad k = 1, \dots, 14. \quad (49)$$

Direct computation gives $\text{Tr}(H_{\text{wind}}) = 0$, $\text{Tr}(H_{\text{wind}}^2) = 80$, $\text{Tr}(H_{\text{wind}}^3) = 0$, $\text{Tr}(H_{\text{wind}}^4) = 2176$. Applying Newton's identities:

$$\begin{aligned} a_1 &= -p_1 = 0, \\ 2a_2 &= -(p_2 + a_1p_1) = -80, \quad a_2 = -40, \\ 3a_3 &= -(p_3 + a_1p_2 + a_2p_1) = 0, \quad a_3 = 0, \\ 4a_4 &= -(p_4 + a_1p_3 + a_2p_2 + a_3p_1) = -(2176 - 3200) = 1024, \quad a_4 = 256. \end{aligned}$$

The higher coefficients $a_5, \dots, a_{14}$ vanish because the rank of H_{wind} is 4 (the nullity is 10). The quartic $\lambda^4 - 40\lambda^2 + 256$ factors as $(\lambda^2 - 8)(\lambda^2 - 32)$. $\square$

5.2 The winding classes and the processor

Definition 5.4 (Winding class partition). The 14 modes partition into three winding classes:

$$A = \{\nu, e, t\}, \quad |A| = 3, \quad (50)$$

$$B = \{u, d, s, \mu, c, \tau, b\}, \quad |B| = 7, \quad (51)$$

$$C = \{\text{dark}, W, Z, H\}, \quad |C| = 4. \quad (52)$$

The partition is the unique one in which each class is homogeneous in fermion/boson content under the processor T .

Definition 5.5 (Processor). The processor T is the rank-10 orthogonal projection

$$T = I - P_C, \quad (53)$$

with P_C the diagonal projector onto winding class C .

Proposition 5.6 (Fermion/boson split). *The processor T commutes with the class- C projector P_C and satisfies*

$$T = I - P_C = P_A + P_B, \quad \text{im}(T) = F = A \cup B, \quad \ker(T) = C. \quad (54)$$

Proof. The processor T is diagonal in the mode basis, and its nonzero entries are exactly on the modes in classes A and B . The class projectors P_A, P_B, P_C are diagonal, orthogonal, idempotent, and sum to the identity. Hence $T = I - P_C = P_A + P_B$. $\square$

The fermion and boson sector dimensions are

$$|F| = d(d+1)/2 = 10, \quad |B| = d = 4. \quad (55)$$

The internal ratios are

$$\frac{|F| - |B|}{|B|} = \frac{d-1}{2} = \frac{3}{2}, \quad \frac{|F|}{|B|} = \frac{d+1}{2} = \frac{5}{2}. \quad (56)$$

5.3 The Kernel-Diagonal Theorem

Theorem 5.7 (Kernel-Diagonal). *Let $P_{\ker}$ be the orthogonal projector onto $\ker(H_{\text{wind}})$. The diagonal of $P_{\ker}$ in the standard mode basis takes exactly two values:*

$$\text{diag}(P_{\ker})_i = \begin{cases} \frac{2}{d} = \frac{1}{2}, & i \in \mathcal{O}_0, \\ \frac{d-1}{d} = \frac{3}{4}, & i \in \mathcal{O}_1 \cup \mathcal{O}_2. \end{cases} \quad (57)$$

Proof. By HCSM-14 Theorem 20, $\{H_{\text{wind}}, P_F\} = 0$, where $F : (n_1, n_2) \mapsto (n_2, n_1)$ is the diagonal flip. Squaring gives $H_{\text{wind}}^2 P_F = P_F H_{\text{wind}}^2$, so P_F commutes with the spectral projector of H_{wind}^2 at eigenvalue 0. Since $\ker(H_{\text{wind}}) = \ker(H_{\text{wind}}^2)$, this gives $[P_{\ker}, P_F] = 0$.

The involution F partitions the 14-mode multiplet into F -cycles. The six F -cycles are

$$\{\nu, e\}, \quad \{u, s\}, \quad \{d, b\}, \quad \{\mu, c\}, \quad \{\text{dark}, \tau\}, \quad \{W, t\}. \quad (58)$$

Since $[P_{\ker}, P_F] = 0$, the diagonal of $P_{\ker}$ is constant on each F -cycle. Direct numerical evaluation gives $(P_{\ker})_{ii} = 1/2$ on $\{\nu, e\} \subset \mathcal{O}_0$ and $3/4$ on the remaining five cycles, which all sit in $\mathcal{O}_1 \cup \mathcal{O}_2$. $\square$

Corollary 5.8 (Three coefficients from one diagonal). *With $P_{\ker}$ as in Theorem 5.7,*

$$c = \frac{d+1}{\text{Tr}(P_{\ker}|_{\mathcal{O}_1 \cup \mathcal{O}_2})} = \frac{5}{9}, \quad (\text{EFT coefficient}), \quad (59)$$

$$\frac{\text{diag}(P_{\ker})|_{\mathcal{O}_1 \cup \mathcal{O}_2}}{\text{diag}(P_{\ker})|_{\mathcal{O}_0}} = \frac{d-1}{2} = \frac{3}{2}, \quad (\text{mass-map coefficient}), \quad (60)$$

$$\dim \ker(H_{\text{wind}}) = 1 + (d-1)^2 = 10, \quad (\text{kernel dimension}). \quad (61)$$

The three coefficients are traces of the same diagonal: the partial trace on $\mathcal{O}_0$ gives $1 = d/d$; the partial trace on $\mathcal{O}_1 \cup \mathcal{O}_2$ gives $(d-1)^2 = 9$; the total trace gives 10.

Proposition 5.9 (F -invariance). $[P_{\ker}, P_F] = 0$, while $[P_{\ker}, R(r)] \neq 0$ and $[P_{\ker}, R(s)] \neq 0$. At machine precision, $\|[P_{\ker}, P_F]\|_F < 10^{-14}$ and $\|[P_{\ker}, R(r)]\|_F = \|[P_{\ker}, R(s)]\|_F = \sqrt{2}$.

Proposition 5.10 (Kernel- T decomposition). $\dim(\ker(H_{\text{wind}}) \cap \text{im}(T)) = 6 = |F| - |B|$ and $\dim(\ker(H_{\text{wind}}) \cap \ker(T)) = 4 = |B|$. The two dimensions sum to $10 = \dim \ker(H_{\text{wind}})$.

5.4 The five-layer loop

The five-layer loop acts diagonally on the 14-mode multiplet:

$$M_{\text{loop}} = T \cdot L_{\text{EFT}} \cdot L_{\text{Core}} \cdot L_{\text{DME}} \cdot L_{\text{FDLM}}. \quad (62)$$

The layers are the FDLM/WIN winding phase (L1), the DME time quantum (L2), the Core gradient flow (L3), the EFT damped mode equation (L4), and the processor T (L5).

Theorem 5.11 (Loop-pass duration). *The loop-pass duration is selected by the deterministic Landauer bound:*

$$T_{\text{pass}} = T_{\text{DME}} = \frac{1}{2\gamma} = 0.7114000035169658 \dots \quad (63)$$

Proof. The processor T takes the maximally mixed state on the 56-dimensional sector ($S_{\text{in}} = \log 56$) to the factorized state on $10 + 4$ modes ($S_{\text{out}} = \log 40$), with entropy reduction

$$\Delta S = \log 56 - \log 40 = \log(7/5) = 0.33647 \dots \text{ nats}. \quad (64)$$

Landauer's principle requires that this entropy reduction be compensated by heat dissipation. Among the HCSM-native scales, $T_{\text{DME}} = 1/(2\gamma)$ is the unique dissipative scale. The coherent beat time $T_{\text{beat}} = \pi/\sqrt{2}$ carries no entropy transfer and is rejected. $\square$

Theorem 5.12 (Exact ratio). $|M_A|/|M_B| = 3 = d - 1$.

Proof. The loop eigenvalue magnitudes are

$$|M_A| = \frac{|w+1|}{2\delta_{\text{edge}}} \exp\left(-\frac{\zeta H}{\rho} \cdot \frac{T_{\text{beat}}}{2}\right), \quad (65)$$

$$|M_B| = \frac{|w+1|}{6\delta_{\text{edge}}} \exp\left(-\frac{\zeta H}{\rho} \cdot \frac{T_{\text{beat}}}{2}\right). \quad (66)$$

The ratio $|M_A|/|M_B| = 6/2 = 3$. All constants cancel. $\square$

5.5 The zero state

The zero state s is the unique nullspace vector of a 14×19 measurement matrix M of vacuum-native observables. The matrix has rank 13, nullity 1, so s is unique up to scale. The smallest singular value of M is 8.47×10^{-16} , machine zero.

Theorem 5.13 (Zero-state structure). *The zero state satisfies*

$$s(rs(m)) = -s(m) \quad \text{for all modes } m, \quad (67)$$

where rs is the diagonal reflection $(n_1, n_2) \mapsto (n_2, n_1)$. The two rs -fixed modes $W = (2, 2)$ and $t = (6, 6)$ have $s(W) = s(t) = 0$ exactly. The six independent amplitudes satisfy the exact ratios $a_3 = 2a_4$ and $a_5 = (2 - \sqrt{2})a_4$. Weight fractions: 3.3% on $\mathcal{O}_0$, 86.5% on $\mathcal{O}_1$, 10.2% on $\mathcal{O}_2$.

Proof. Direct computation of the nullspace via singular value decomposition of M , then verification of the symmetry $s(rs(m)) = -s(m)$ and the algebraic ratios at machine precision. $\square$

6 Level 4: Output Structures

6.1 The charge functional

The electric charge in units of $e/3$ is given by the functional

$$q_3 = F(n_1, \sigma), \quad \sigma = \text{sign}(s), \quad (68)$$

where n_1 is the mode's first momentum index and s is the zero-state amplitude.

Theorem 6.1 (Charge bijection). *The 14 modes have 14 pairwise distinct (n_1, σ) tuples. The map F is well-defined on the realized pairs and is a bijection onto the five Standard Model charge eigenvalues $\{-3, -1, 0, +2, +3\}$.*

Theorem 6.2 (σ -line sums). *The charge sums over the three σ -lines are*

$$\sum_{\sigma=-1} q_3 = 0, \quad \sum_{\sigma=0} q_3 = +5, \quad \sum_{\sigma=+1} q_3 = -8. \quad (69)$$

The total vacuum charge is $\sum_i q_{3i} = -3 = -1e$.

6.2 The mass map

The mass map is

$$m_i = A_{\text{mass}} \exp(-B_{\text{mass}}(N/4)_i), \quad (70)$$

with

$$A_{\text{mass}} = \frac{(2d-1)N}{2} = 224 \text{ GeV}, \quad B_{\text{mass}} = \frac{(d-1)kL^2}{2N^2} = 0.5411967 \dots \quad (71)$$

The map reproduces the twelve charged fermion masses and the W , Z , H , and t masses at RMS 0.0449 dex across thirteen orders of magnitude in mass.

6.3 Three derivations of the coefficient $3/2$

The coefficient $(d-1)/2 = 3/2$ has three independent derivations.

1. **Processor F/B split.** From the fermion and boson sector dimensions,

$$\frac{|F| - |B|}{|B|} = \frac{10 - 4}{4} = \frac{3}{2} = \frac{d-1}{2}. \quad (72)$$

2. **DNLS Hartree orbit trace.** From the DNLS Hartree kernel,

$$\frac{\text{Tr } K_{\mathcal{O}_1}}{\text{Tr } K_{\mathcal{O}_0}} = \frac{3}{2}. \quad (73)$$

3. **Kernel-diagonal ratio.** From Corollary 5.8,

$$\frac{\text{diag}(P_{\text{ker}})|_{\mathcal{O}_1 \cup \mathcal{O}_2}}{\text{diag}(P_{\text{ker}})|_{\mathcal{O}_0}} = \frac{3/4}{1/2} = \frac{3}{2}. \quad (74)$$

Derivation (3) is the most primitive: it uses only the projector onto $\ker(H_{\text{wind}})$.

7 Level 5: Emergent Physics

7.1 Gauge algebra from the D_4 commutant

Theorem 7.1 (Sector-coarse correspondence). *The D_4 commutant on the complexified 14-mode multiplet $V_{14} = \mathbb{C}^{14}$ has complex dimension 27 and decomposes as*

$$\text{Comm}(D_4, V_{14}) \cong M_3(\mathbb{C}) \oplus M_1(\mathbb{C}) \oplus M_2(\mathbb{C}) \oplus M_2(\mathbb{C}) \oplus M_3(\mathbb{C}). \quad (75)$$

The gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$ embeds in this commutant as a Lie subalgebra of real dimension 16: $\mathfrak{su}(3)$ in the A_1 block, $\mathfrak{su}(2)$ in the B_1 block, and $\mathfrak{u}(1)^5$ spanned by the five isotypic projectors.

Proof. By Schur's lemma, the commutant of a representation $V = \bigoplus_{\rho} V_{\rho} \otimes M_{\rho}$ is $\bigoplus_{\rho} \text{End}(M_{\rho}) = \bigoplus_{\rho} M_{m_{\rho}}(\mathbb{C})$. The isotypic decomposition of V_{14} is $3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E$, giving multiplicities $(3, 1, 2, 2, 3)$ and $\sum m_{\rho}^2 = 9 + 1 + 4 + 4 + 9 = 27$.

The $\mathfrak{su}(3)$ subalgebra is the traceless anti-Hermitian part of the A_1 block $M_3(\mathbb{C})$, of real dimension 8. The $\mathfrak{su}(2)$ subalgebra is the traceless anti-Hermitian part of the B_1 block $M_2(\mathbb{C})$, of real dimension 3. The $\mathfrak{u}(1)^5$ subalgebra is spanned by the five isotypic projectors, of real dimension 5. These three factors close to $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$, of real dimension $8 + 3 + 5 = 16$. $\square$

The A_1 factor is identified with physical color, and the B_1 factor with physical weak isospin.

7.2 Hypercharge quantization

Theorem 7.2 (Hypercharge). *Matching the Standard Model hypercharges $Y(u_R^c) = -2/3$ on Ω^0 , $Y(Q_L) = +1/6$ on Ω^1 , $Y(d_R^c) = +1/3$ on Ω^2 on the A_1 isotypic block, and matching $Y(e_R^c) = +1$ on Ω^0 , $Y(L_L) = -1/2$ on Ω^1 , $Y(\nu_R^c) = 0$ on Ω^2 on the B_1 block, the hypercharge functional takes the unique coefficients*

$$(\alpha_{A_1}, \beta_{A_1}, \gamma_{A_1}) = (0, -1/6, -1/2), \quad (76)$$

$$(\alpha_{B_1}, \beta_{B_1}, \gamma_{B_1}) = (0, +1/2, +1/2), \quad (77)$$

giving hypercharge values in $(1/6)\mathbb{Z}$ on the three Hodge sectors.

Proof. On the A_1 block, matching the three sector values gives the linear system

$$\alpha_{A_1} + \beta_{A_1} + \gamma_{A_1} = -2/3,$$

$$\alpha_{A_1} - \beta_{A_1} = +1/6,$$

$$\alpha_{A_1} + \beta_{A_1} - \gamma_{A_1} = +1/3.$$

The 3×3 coefficient matrix has determinant 4, so the solution is unique. Adding the first and third equations gives $2\alpha_{A_1} + 2\beta_{A_1} = -1/3$, so $\alpha_{A_1} + \beta_{A_1} = -1/6$. Substituting $\alpha_{A_1} = \beta_{A_1} + 1/6$ gives $2\beta_{A_1} = -1/3$, so $\beta_{A_1} = -1/6$, $\alpha_{A_1} = 0$, $\gamma_{A_1} = -1/2$. The B_1 system is solved the same way. $\square$

7.3 Flavor: the CKM matrix

The Cabibbo angle satisfies

$$\sin \theta_{12} = \frac{\pi}{14} = 0.2243995 \dots \quad (78)$$

The winding amplitude $a = 3/5$ is topological: it is the ratio $(|F| - |B|)/|F| = 6/10$ of exact rank counts, protected from δ_{edge} corrections. The Wolfenstein parameters are

$$\lambda = \frac{\pi}{14}, \quad A = \frac{d}{d+1}(1 + \delta_{\text{edge}}), \quad \sqrt{\rho^2 + \eta^2} = \frac{d+1}{d(d-1)} = \frac{5}{12}. \quad (79)$$

7.4 Flavor: the PMNS matrix

The neutrino $N/4$ spectrum is $(60, 60, 58)$. The PMNS magnitudes match at RMS 0.035. The mixing angles are $\theta_{12} = 34.2^\circ$, $\theta_{23} = 47.9^\circ$, $\theta_{13} = 8.5^\circ$, and the Dirac CP phase is $\delta_{CP} \approx 197^\circ$. The generation phase

$$\phi_{\text{gen}} = \frac{|F| - |B|}{|F|} \pi = \frac{3}{5} \pi \quad (80)$$

is exact by the topological protection theorem. Majorana phases are $(\alpha_1, \alpha_2) = (\pi, 0) \bmod \pi$.

7.5 Higgs sector

The top-Yukawa coefficient is

$$y_t^2 = 1 - \frac{4}{3} \delta_{\text{edge}}, \quad \frac{4}{3} = \frac{|\mathcal{O}_2|}{d-1}. \quad (81)$$

The top mass is $m_t = 172.688398$ GeV (PDG 172.690 ± 0.30 GeV). The tree-level quartic coupling is $\lambda_{\text{tree}} = \ln(kL)/[(d-1)\pi^2]$, and the electroweak vacuum expectation value is $v_{\text{EW}} = M_P \exp(-kL_{\text{phys}})$.

7.6 Gravity

The graviton is the self-paired $B_1 \oplus B_2$ curvature mode of $\text{Sym}^2(\Omega^1|_{\lambda=4})$, consisting of $54B_1 \oplus 54B_2 = 108$ modes. It is exactly massless at all orders by D_4 Schur orthogonality. The Newton constant is

$$G_N = \frac{(d+1)^2}{8\pi M_P^2} = \frac{25}{8\pi M_P^2}. \quad (82)$$

Masslessness by Schur. The graviton lives in the $B_1 \oplus B_2$ isotypic component of the symmetric square of $\Omega^1|_{\lambda=4}$. Any mass term invariant under D_4 must be proportional to the identity on each isotypic component, but the $B_1 \oplus B_2$ component carries no D_4 -invariant mass operator. Hence the graviton remains massless. $\square$

7.7 Absence of a light dark photon

Theorem 7.3. *The kinetic mixing parameter satisfies $\epsilon \equiv 0$ exactly over all mass scales.*

Proof. The HCSM gauge algebra $\mathfrak{su}(3)_c \oplus \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^5$ contains no hidden Abelian factor coupling to electrically charged matter. The hidden sector $5+6+31$ is $U(1)_Y$ -neutral. Consequently no tree-level kinetic mixing operator $F_{\mu\nu} F'^{\mu\nu}$ can be written, no one-loop kinetic mixing is generated, and no state is charged under both the visible $\mathfrak{u}(1)_{\text{em}}$ and any hidden Abelian factor. $\square$

7.8 Spin chirality $\mathbb{Z}_2$ grading

The 56-dimensional Hodge lift at $\lambda = 4$ carries three commuting $\mathbb{Z}_2$ operators: $\gamma_5 = P_{\Omega^0} - P_{\Omega^1} + P_{\Omega^2}$ (Hodge form-parity), χ_{spin} (spin-structure parity), and $F : (n_1, n_2) \mapsto (n_2, n_1)$ (orientation flip). All three square to I and pairwise commute, generating $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$. Consequences: spin-statistics from γ_5 with $\{\gamma_5, D\} = 0$ and $\text{Tr}(\gamma_5 f(D)) = 0$ for any even regulator f ; chirality from χ_{spin} ; CP from F .

7.9 Strong CP from orientation $\mathbb{Z}_2$

Theorem 7.4. $\theta_{\text{QCD}} = \theta_{\text{eff}} = 0$ exactly.

Proof. On the 14-mode multiplet at $\lambda = 4$, $[F, \Delta] = 0$ and $\{F, H_{\text{wind}}\} = 0$ exactly, while $[F, H_{\text{wind}}] \neq 0$. The winding matrix M is antisymmetric under F ; the odd function $\sin(\pi M/8)$ preserves the antisymmetry; hence $P_F H_{\text{wind}} P_F^{-1} = -H_{\text{wind}}$. The physical identification $H_{\text{wind}} \leftrightarrow F_{\mu\nu} \tilde{F}^{\mu\nu}$ gives $\theta_{\text{eff}} = 0$ exactly. $\square$

7.10 Proca sector and $\mathfrak{u}(1)_Y$ origin

The $\mathfrak{u}(1)_Y$ gauge symmetry is generated by the hypercharge functional $\hat{Y}$ on $\Omega^0|_{\lambda=4}$: $[\hat{Y}, T] = 0$, $[\hat{Y}, F] = 0$, eigenvalues in $(1/6)\mathbb{Z}$. The Proca Lagrangian is

$$\mathcal{L}_{\text{Proca}} = -\frac{1}{4}W_{\mu\nu}^a W^{a\mu\nu} - \frac{1}{4}B_{\mu\nu} B^{\mu\nu} + \frac{1}{2}m_W^2 W_\mu^+ W^{-\mu} + \frac{1}{2}m_Z^2 Z_\mu Z^\mu, \quad (83)$$

with $m_W = gv_{\text{EW}}/2$ and $m_Z = \sqrt{g^2 + g'^2} v_{\text{EW}}/2$. The mass ratio $m_W/m_Z = \cos \theta_W = 0.8769$ matches PDG 0.8814 at 0.5%. Tree-level $\rho = 1$ exactly.

8 Level 6: Cosmology

8.1 The kL crossover

The bare warp factor is exact:

$$kL_{\text{bare}} = \frac{(d-1)N}{d+1} = \frac{3 \cdot 64}{5} = \frac{192}{5} = 38.4. \quad (84)$$

The physical warp factor is $kL_{\text{phys}} = 38.442527$, matching $\ln(M_P/v_{\text{EW}})$ to 0.004%. The correction is

$$kL_{\text{phys}} - kL_{\text{bare}} = \frac{|D_4| - 1}{2} \delta_{\text{edge}} = \frac{7}{2} \delta_{\text{edge}}. \quad (85)$$

8.2 Dark energy and the DME entropy-well normalization

The bare dark energy density is

$$\rho_{\text{DE}}^{\text{bare}} = \frac{|F|}{|B|} m_\nu^4 = \frac{d+1}{2} m_\nu^4 = 2.4809 \times 10^{-47} \text{ GeV}^4, \quad (86)$$

matching observation to 0.76%. The refined formula is

$$\rho_{\text{DE}} = \frac{d+1}{2} (1 + c\delta) m_\nu^4, \quad c = \frac{d+1}{(d-1)^2} = \frac{5}{9}, \quad \delta = \frac{(1-\gamma)^2}{2\pi}, \quad (87)$$

giving $\rho_{\text{DE}} = 2.5003 \times 10^{-47} \text{ GeV}^4$, matching observation to 0.010%.

The DME entropy-well normalization is

$$S_0 = v_{\text{EW}}^2 \cdot \frac{7}{40} \cdot \left(1 + \frac{\delta_{\text{edge}}}{6}\right) = 10630.722 \text{ GeV}^2, \quad (88)$$

matching the implied value at relative residual 2.44×10^{-6} (18.8% of the M_P budget).

8.3 Vacuum stability

The stability coefficient is

$$\gamma = \frac{c_2}{I(2)} = 0.702839467990 \dots \quad (89)$$

with

$$I(2) = 1 - \frac{1}{2} \text{AGM}(1, \sqrt{2}), \quad (90)$$

$$\text{AGM}(1, \sqrt{2}) = \frac{1}{\sqrt{2} \eta(i)^2}, \quad (91)$$

$$\eta(i) = \frac{\Gamma(1/4)}{2\pi^{3/4}}. \quad (92)$$

The stability theorem states $S''(0)/S_0 = -2\gamma < 0$: the equilibrium $D = 4$ is a strict local maximum of the vacuum entropy.

8.4 The Λ -promotion mechanism

The residual vacuum energy after mirror cancellation is promoted to the observed cosmological constant by pure projection, not by gravitational coupling.

The five-layer loop $M_{\text{loop}} = T \cdot L_{\text{EFT}} \cdot L_{\text{Core}} \cdot L_{\text{DME}} \cdot L_{\text{FDLM}}$ acts on the 14-mode multiplet. The EFT layer L_{EFT} acts as a soft projector with weight $1 + c\delta$, where $c = 5/9$ is set by the Kernel-Diagonal Theorem. The residual energy projects onto the timelike direction $(v_0 - v_2)/\sqrt{2}$ of $\ker D$, with unit projection weight (the residual is a scalar). The hidden $\mathfrak{u}(1)$ factors contribute zero: their generators are traceless on each D_4 multiplicity space of the hidden sector, and the hidden vacuum is the D_4 -invariant nullspace of the hidden measurement matrix, so all hidden $\mathfrak{u}(1)$ VEVs vanish at machine precision.

The mechanism is therefore a projection, not a coupling. There is no gravitational suppression factor $1/M_P^2$; the residual vacuum energy is already at the right scale because it is topological. Combined with the absence of a light dark photon (Theorem 7.3), this confirms that no gauge mediator exists between the hidden and visible sectors, and the only route for the residual energy to affect the visible sector is universal gravitation via the projection onto $\ker D$. This closes item 6.13 of the framework's need-to-derive list.

9 Falsifiable predictions

The framework makes several predictions that distinguish it from generic Standard Model extensions.

Falsifiable Prediction 9.1 (No light dark photon). The kinetic mixing parameter satisfies $\epsilon \equiv 0$ exactly. The prediction is falsifiable by SHiP, FASER2, MATHUSLA, CODEX-b, LDMX, NA64, Belle II, and HL-LHC at $\epsilon > 10^{-12}$.

Falsifiable Prediction 9.2 (Strong CP). $\theta_{\text{QCD}} = 0$ exactly. Any non-zero measurement of the neutron electric dipole moment at the level predicted by a generic θ_{QCD} would falsify the framework.

Falsifiable Prediction 9.3 (ρ parameter). Tree-level $\rho = m_W^2/(m_Z^2 \cos^2 \theta_W) = 1$ exactly. Any measured deviation of ρ from 1 at tree level would falsify the framework.

Falsifiable Prediction 9.4 (Fermion content). One Standard Model generation contains exactly 14 particles. A fourth generation would falsify the framework.

Falsifiable Prediction 9.5 (Kernel structure). The winding kernel has dimension 10 on Λ_8 and the kernel-diagonal values are $\{1/2, 3/4\}$. Any modification of the spectrum of H_{wind} that changes these values would falsify the framework.

Falsifiable Prediction 9.6 (Dark energy). The dark energy density is $\rho_{\text{DE}} = \frac{d+1}{2}(1 + c\delta)m_\nu^4$, matching observation to 0.010%. Any future measurement of ρ_{DE} that deviates from this formula at more than 0.1% would falsify the framework.

10 Theorem inventory

Theorem	Location
Laplacian uniqueness	HCSM-01 Theorem 2.2.1
$d \cdot 2^d = 64$, $L = 8$	HCSM-01 Consistency 4.1.1
Two derivations of $L = 8$	HCSM-01 §4, HCSM-14.b
$2(L - 1) = 14$	HCSM-01 Theorem 4.5.1
Hodge–Dirac square	HCSM-02 Theorem 3.2.1
Supercharge algebra	HCSM-02 Theorem 4.3.1
Kernel dimension = 4	HCSM-02 Theorem 6.1.1
Spectral gap $2 \sin(\pi/L)$	HCSM-02 Theorem 6.2.1
Lorentzian metric conditional uniqueness	HCSM-03 Theorem 5.2.1
Self-paired eigenvalue $\lambda = 4$	HCSM-04 Theorem 3.2
D_4 orbit decomposition $(2, 8, 4)$	HCSM-04 Theorem 5.1
Support classes $\{64, 48, 32\}$	HCSM-04 Theorem 6.2
IPRs $\{2/128, 3/128, 4/128\}$	HCSM-04 Theorem 7.2
Entropy classes $\{6.0, 5.5, 5.0\}$	HCSM-04 Theorem 8.2
Sector-coarse gauge algebra	HCSM-06 Theorem 5.10
Hypercharge quantization	HCSM-07 Theorem 7.1
Orientation-flip $\mathbb{Z}_2$	HCSM-09 Theorem 5.1
Winding class partition $(3, 7, 4)$	HCSM-14 Theorem 15
Kernel-Diagonal Theorem	HCSM-14.a Theorem 1
Three coefficients from one diagonal	HCSM-14.a Corollary 2
F -invariance of kernel	HCSM-14.a Theorem 4
Kernel- T decomposition	HCSM-14.a Theorem 5
Maximal cancellation selects $L = 8$	HCSM-14.b Theorem 4.4.1
Loop-pass duration $T_{\text{DME}} = 1/(2\gamma)$	HCSM-21 Theorem 12.1
Exact ratio $ M_A / M_B = 3$	HCSM-20 Theorem 9.4
Charge bijection	HCSM-24 Theorem 4.1
σ -line sums	HCSM-24 Theorem 5.1
Top-Yukawa coefficient $4/3$	HCSM-40
Absence of light dark photon	HCSM-41 Theorem 5
Spin chirality $\mathbb{Z}_2$ grading	HCSM-42
Strong CP $\theta_{\text{QCD}} = 0$	HCSM-43 Theorem 6
Proca sector	HCSM-44
Graviton masslessness by Schur	HCSM-45, 46
$G_N = (d + 1)^2/(8\pi M_P^2)$	HCSM-48
kL correction $(7/2)\delta_{\text{edge}}$	HCSM-51
EFT coefficient $c = 5/9$	HCSM-52
S_0 normalization	HCSM-53
Vacuum stability $\gamma = c_2/I(2)$	HCSM-54
Λ -promotion is a projection	HCSM-56.a

All seven rank blockers of the framework’s derivation program are closed:

Rank	Blocker	Status	Paper
1	HCSM-native γ	Closed	HCSM-54
2	kL bare	Closed	HCSM-51
3	Charge bijection	Closed	HCSM-24
4	$N = d \cdot 2^d$	Closed	HCSM-01
5	$(d-1)/2$ from DNLS	Closed	HCSM-19
6	3D irrep for CKM	Closed by reframing	HCSM-34
7	Loop-pass duration	Closed	HCSM-21

11 Reproducibility

All numerical claims in the HCSM paper stack are verified at machine precision in IEEE-754 double precision. The verification scripts use only standard libraries (NumPy, SciPy, SymPy, mpmath) and construct the objects they need from HCSM-native definitions. The verification scripts are indexed by paper number; running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the stack.

The complete HCSM paper stack, the verification scripts, and the $\text{\LaTeX}$ and TXT source of each paper are available at

<https://github.com/007STAN/HODGE-COMPLEX-STANDARD-MODEL-HCSM->

12 Conclusion

This paper has established the foundation of the Hodge Complex Standard Model. The substrate is the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$, selected by two independent derivations: equal entropy spacing on the torus Laplacian, and maximal cancellation of the winding generator at $L = 8$. The substrate carries the Hodge complex $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$ with $D^2 = \Delta$ exactly on each form degree. The 14 modes at the self-paired eigenvalue $\lambda = 4$ are identified with one Standard Model generation. The processor T and its five-layer loop are the machine's operation. The output structures (charge, mass, flavor, gravity, cosmology) are the residue after mirror cancellation.

The framework's most consequential structural results are three. First, the Kernel-Diagonal Theorem unifies the framework's three leading coefficients under a single projector: the EFT coefficient $c = 5/9$, the mass-map coefficient $(d-1)/2 = 3/2$, and the kernel dimension $10 = (d-1)^2 + 1$ are three traces of the diagonal of the projector onto $\ker(H_{\text{wind}})$. Second, the Λ -promotion mechanism is closed: the residual vacuum energy is promoted by pure projection onto the timelike direction of $\ker D$, and the hidden $\mathfrak{u}(1)$ VEVs vanish. Third, the framework's coverage of the Standard Model gauge content (absence of a light dark photon, spin-chirality $\mathbb{Z}_2$ grading, strong-CP solution, Proca sector, topological protection of ϕ_{gen}) is complete.

The framework has a single dimensionful anchor (the Planck mass M_P) and derives every other quantity from the substrate at $d = 4$ and $L = 8$.

The framework's claim is that the Standard Model is what the substrate looks like when you observe it. The Hodge complex is the machine. Physics is what the machine cannot cancel.

References

- [1] S. Preschutti, *HCSM-01: The 8×8 Torus Substrate from $d = 4$* , HCSM White Paper Series (2026).
- [2] S. Preschutti, *HCSM-02: The Hodge Complex on Λ* , HCSM White Paper Series (2026).

- [3] S. Preschutti, *HCSM-03: Lorentzian Signature from the Hodge Star*, HCSM White Paper Series (2026).
- [4] S. Preschutti, *HCSM-04: The 14-Mode Multiplet at $\lambda = 4$* , HCSM White Paper Series (2026).
- [5] S. Preschutti, *HCSM-05: Orbit Decomposition and Inverse Participation*, HCSM White Paper Series (2026).
- [6] S. Preschutti, *HCSM-06: Sector-Coarse Correspondence*, HCSM White Paper Series (2026).
- [7] S. Preschutti, *HCSM-07: Hypercharge Quantization*, HCSM White Paper Series (2026).
- [8] S. Preschutti, *HCSM-09: Gauge Topology and Orientation-Flip $\mathbb{Z}_2$* , HCSM White Paper Series (2026).
- [9] S. Preschutti, *HCSM-13: The Processor T and the Fermion/Boson-Class Correspondence*, HCSM White Paper Series (2026).
- [10] S. Preschutti, *HCSM-14: Winding Generator Spectrum and Class Partition*, HCSM White Paper Series (2026).
- [11] S. Preschutti, *HCSM-14.a: The Kernel-Diagonal Theorem*, HCSM White Paper Series (2026).
- [12] S. Preschutti, *HCSM-14.b: Maximal Cancellation Selects $L = 8$* , HCSM White Paper Series (2026).
- [13] S. Preschutti, *HCSM-19: DNLS Hartree Mass-Map Coefficient*, HCSM White Paper Series (2026).
- [14] S. Preschutti, *HCSM-20: The Motion Theorem*, HCSM White Paper Series (2026).
- [15] S. Preschutti, *HCSM-21: The Time Theorem*, HCSM White Paper Series (2026).
- [16] S. Preschutti, *HCSM-22: The Zero State*, HCSM White Paper Series (2026).
- [17] S. Preschutti, *HCSM-23: Identification of the 14 Modes with One Standard Model Generation*, HCSM White Paper Series (2026).
- [18] S. Preschutti, *HCSM-24: The Charge Theorem*, HCSM White Paper Series (2026).
- [19] S. Preschutti, *HCSM-26: The Observer Pair $\{W, t\}$* , HCSM White Paper Series (2026).
- [20] S. Preschutti, *HCSM-33: Mass Map*, HCSM White Paper Series (2026).
- [21] S. Preschutti, *HCSM-34: The CKM Matrix*, HCSM White Paper Series (2026).
- [22] S. Preschutti, *HCSM-35: The Daughter Structure of the CKM*, HCSM White Paper Series (2026).
- [23] S. Preschutti, *HCSM-36: The PMNS Matrix*, HCSM White Paper Series (2026).
- [24] S. Preschutti, *HCSM-37: PMNS Generation Phase*, HCSM White Paper Series (2026).
- [25] S. Preschutti, *HCSM-40: The Higgs Mechanism*, HCSM White Paper Series (2026).
- [26] S. Preschutti, *HCSM-41: Absence of a Light Dark Photon*, HCSM White Paper Series (2026).

- [27] S. Preschutti, *HCSM-42: Spin Chirality $\mathbb{Z}_2$ Grading*, HCSM White Paper Series (2026).
- [28] S. Preschutti, *HCSM-43: Strong CP from the Orientation $\mathbb{Z}_2$* , HCSM White Paper Series (2026).
- [29] S. Preschutti, *HCSM-44: The Proca Sector*, HCSM White Paper Series (2026).
- [30] S. Preschutti, *HCSM-45: Graviton as Self-Paired Curvature Mode*, HCSM White Paper Series (2026).
- [31] S. Preschutti, *HCSM-48: Faddeev-Popov G_N Normalization*, HCSM White Paper Series (2026).
- [32] S. Preschutti, *HCSM-51: The kL Crossover*, HCSM White Paper Series (2026).
- [33] S. Preschutti, *HCSM-52: The EFT Coefficient in the Dark Energy Density*, HCSM White Paper Series (2026).
- [34] S. Preschutti, *HCSM-53: Dark Energy Density and the DME Entropy-Well Normalization*, HCSM White Paper Series (2026).
- [35] S. Preschutti, *HCSM-54: Vacuum Stability and the Stability Coefficient*, HCSM White Paper Series (2026).
- [36] S. Preschutti, *HCSM-56.a: Λ -Promotion as Pure Projection*, HCSM White Paper Series (2026).
- [37] J. Dodziuk, *Finite-difference approach to the Hodge theory of harmonic forms*, Amer. J. Math. **98**, 79 (1976).
- [38] M. Nakahara, *Geometry, Topology and Physics*, 2nd ed., CRC Press (2003).
- [39] C. Voisin, *Hodge Theory and Complex Algebraic Geometry I*, Cambridge University Press (2002).
- [40] Particle Data Group, Phys. Rev. D **110**, 030001 (2024).